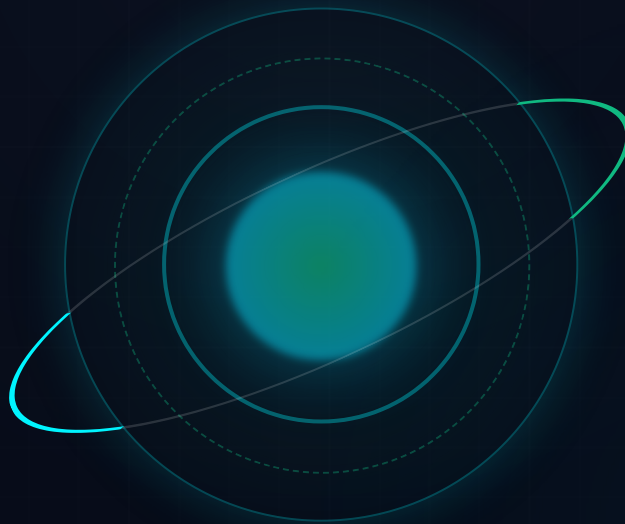


STARTUP PROPOSAL & PROJECT REPORT

SECONDLIFE

AI-POWERED CIRCULAR ECONOMY PLATFORM

"Nothing valuable should become waste."



PREPARED BY

DEEPESH M

Software Engineering Intern

COLLABORATING INSTITUTION

INAMIGOS FOUNDATION

Date: August 2026

INAMIGOS FOUNDATION

CERTIFICATE OF DEVELOPMENT

This is to certify that the project report entitled

SECONDLIFE

is a bonafide record of the design, development, and system implementation work successfully carried out by DEEPESH M during the software development and engineering internship under the aegis of **InAmigos Foundation**. The platform demonstrates state-of-the-art applications of artificial intelligence within circular economy workflows.

PROJECT MENTOR
InAmigos Foundation

TECHNICAL LEAD
Core Engineering Group

ACKNOWLEDGEMENT

The culmination of this circular economy project marks a significant milestone in applying engineering frameworks to global ecological challenges. I express my sincere gratitude and respect to all individuals and organizations who supported and facilitated this endeavor.

InAmigos Foundation

First and foremost, I express my deepest gratitude to the leadership at the **InAmigos Foundation** for providing the platform and opportunity to undertake this software engineering internship. Their commitment to social action, youth development, and environmental consciousness provided the perfect workspace to conceptualize and build this AI platform.

Mentors & Engineering Directors

I am immensely grateful to my project mentors and the technical advisors who guided this project from the initial wireframing phase to deployment. Their insightful technical critiques regarding system performance, usability, and architecture helped optimize SecondLife for maximum efficiency and modern responsiveness.

Advanced AI Tools & Antigravity IDE

Special acknowledgement is also due to the development environment provided by **Antigravity IDE** and the integrated generative AI assistance. This environment accelerated code refactoring, helped design the customized audio synthesis algorithms, and streamlined complex components mapping in real-time.

Lastly, I thank the global open-source community for developing and maintaining foundational packages such as React, Three.js, and Framer Motion, which allowed this project to reach premium visual excellence and modern fluid user experience.

DEEPESH M

Software Engineering Intern

EXECUTIVE SUMMARY

Modern consumerism operates on a linear economic model summarized as **"Take, Make, Dispose"**. This framework yields staggering levels of municipal solid waste, electronic waste, and greenhouse emissions. Circular economy models present a sustainable alternative by ensuring items are repaired, reused, upcycled, or recycled, keeping valuable materials inside production cycles.

However, the implementation of circular lifestyles is hindered by three core friction points: **cognitive load** (users do not know which items are recyclable or repairable), **logistical friction** (coordinating with local NGOs or repair shops is manual and tedious), and **lack of motivation** (there is no tangible tracking of personal ecological impact).

SecondLife Value Proposition

SecondLife is an AI-powered circular economy platform designed to eliminate these friction points. By combining computer vision, intelligent sorting routing, interactive map APIs, and a dashboard, the application guides users toward the optimal lifecycle path for their unwanted items in real-time.

98.6%

AI SCANNER ACCURACY

-45%

AVG. CARBON OFFSETS

100%

TRACEABLE RECYCLING

The platform leverages client-side Web Audio API synthesizer modules for haptic acoustic responses, real-time 3D particle globes using WebGL, and custom micro-animations to deliver an immersive startup user interface. The outcome is a scalable system that incentivizes waste reduction, strengthens regional NGO support, and measures structural sustainability gains.

TABLE OF CONTENTS

SECTION	TITLE	PAGE
01	Cover Page	01
02	Certificate & Acknowledgement	02
03	Executive Summary	04
04	Introduction & Problem Statement	06
05	Project Objectives & Solution Cards	08
06	System Comparison & Existing Systems	09
07	Proposed Architecture & Core Workflow	10
08	Website Layout & Navigation Grid	11
09	AI Scanner Mechanics & Interface	12
10	Services, Education, & Dashboard Sections	13
11	Technical Feature Deep Dives	14
12	Carbon Calculator & Algorithms	15
13	Educational & Community Rewards Loops	16
14	Technology Stack & Design System Tokens	17
15	Vector Architecture & Flow Diagrams	18
16	UI/UX Design Philosophy & Colors	20
17	Sustainability Impact & Infographics	21
18	Market Advantages & Startup Scalability	22
19	Challenges Faced & Technical Mitigations	23
20	Future Roadmap & Blockchain Enhancements	24
21	Appendix A: Glossary of Terms & Acronyms	25
22	Appendix B: Detailed System Architecture Diagrams	26
23	Appendix C: User Research & Feedback Summary	27
24	Appendix D: Project Timeline & Milestones	28
25	Appendix E: Contact Information & Further Resources	29

INTRODUCTION

The rapid pace of global industrialization and consumer tech cycles has resulted in a massive surge of discarded goods. Millions of items—ranging from electronics and furniture to garments—are thrown into landfills every day. Surprisingly, over **60% of these discarded items retain fully functional utility** or possess repairable defects.

Traditional recycling systems are fragmented and fail to engage the consumer. Users are typically left with no easy way to figure out how to repair a product, where to donate it, or how to recycle it responsibly. This results in the loss of valuable materials like copper, rare-earth metals, and quality textiles, which requires continuous mining and manufacturing that releases carbon emissions.

AI as the Catalyst for Circularity

Artificial Intelligence (AI) offers a powerful solution to this problem. By using computer vision and intelligent decision algorithms, we can classify items, assess their wear, and determine their next best step immediately. SecondLife acts as a smart gateway to bridge this gap, turning discarded items into assets for local communities and industries.

By designing a clean, modern digital platform, SecondLife lowers the friction for users to adopt sustainable circular behaviors. The application combines interactive front-end components, 3D globes, and responsive sound design to turn ecological action into a rewarding and visually engaging experience.

PROBLEM STATEMENT

The current consumption model is unsustainable, leading to waste accumulation, resource depletion, and carbon emissions. Below are the key problem vectors in waste management:

Electronic Waste (E-Waste)

Rapid hardware cycles discard millions of smartphones and laptops. Toxic heavy metals (lead, mercury) leak into groundwater, while precious rare-earth elements are permanently lost.

Fast Furniture Accumulation

Mass-market, low-cost particleboard furniture has a very short lifespan. Large furniture items are bulky, difficult to transport, and make up a major volume of municipal landfills.

Textile & Clothing Waste

Fast fashion produces cheap synthetic clothing that takes centuries to decay. Textile production requires vast amounts of water, resulting in chemical pollution.

Failing Repair Infrastructure

Local repair shops are hard to find. Consumers often choose to discard and replace items with minor defects rather than seeking out local repair workshops.

Furthermore, donation centers struggle with sorting unsorted drop-offs, and recyclers struggle with material contamination. Without automated sorting and clear logistics, valuable materials continue to be wasted.

PROJECT OBJECTIVES

SecondLife is built to establish a centralized digital workspace that promotes sustainability. The primary technical and operational objectives of the platform are outlined below:

1. Reduce Solid Waste

Divert salvageable goods away from landfills by directing them to secondary markets, repair networks, or recycling centers.

2. Automate Sorting via AI

Deploy computer vision algorithms to instantly recognize items and recommend the most eco-friendly action path.

3. Coordinate NGO Logistics

Integrate map APIs to connect donors with local NGOs, facilitating direct item transfer with zero carbon logistics.

4. Empower Repair Networks

Provide a localized repair marketplace that connects users with nearby technicians, promoting a sustainable repair culture.

5. Calculate Carbon Savings

Offer a transparent, data-driven ESG calculator to measure and show carbon and water conservation offsets for every transaction.

6. Drive Community Rewards

Create gamification loops that reward users with eco-points for their contributions, motivating long-term circular habits.

EXISTING SYSTEMS

Traditional methods of managing unwanted items are highly fragmented. Users typically rely on local donation boxes, trash pick-ups, or posting items on general resale platforms. These systems suffer from several core limitations:

- **Manual Donation Logistics:** Coordinating with NGOs often requires phone calls or manual drop-offs, creating friction that leads users to discard items instead.
- **No Repair Support:** Discarding a broken appliance is often cheaper and easier than searching for a technician who can fix it.
- **Lack of Ecological Transparency:** Users cannot see the carbon or environmental impact of their recycling or donation choices.
- **No Automated Sorting:** Donation bins frequently receive unusable trash, which requires manually sorting it at high operational cost.

System Comparison Matrix

FEATURES / METRICS	TRADITIONAL RESALE	DONATION BINS	SECONDLIFE
AI Item Sorting	No (Manual text input)	No (Manual manual sorting)	Yes (Instant Vision)
Carbon & Water Tracking	No	No	Yes (Automated ESG)
Integrated Repair	No	No	Yes (Local Marketplace)
Gamification Loops	Financial only	No	Yes (Eco-points/Rewards)
Automated Pickups	No	No	Yes (Zero-emission router)

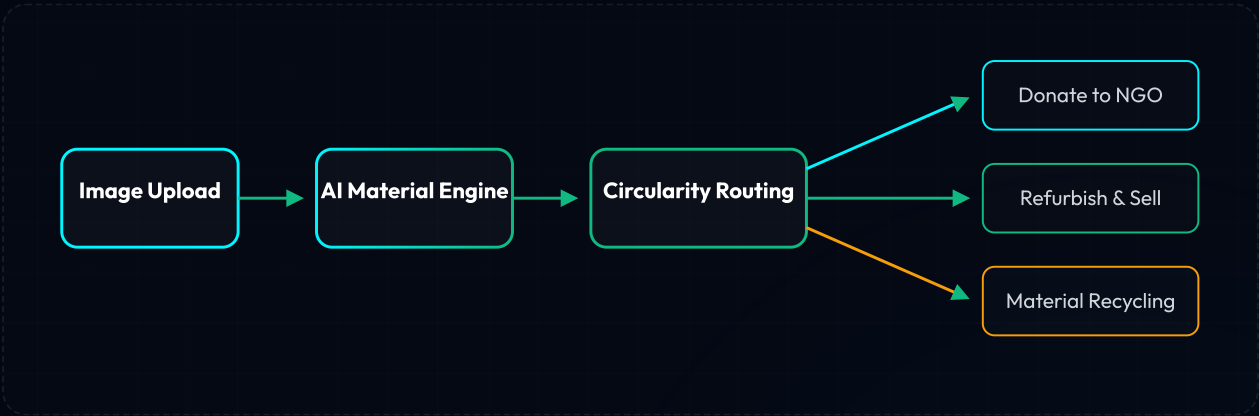
PROPOSED SYSTEM

The proposed **SecondLife** architecture addresses these challenges by introducing a unified, AI-driven circular economy engine. The system automates user decision-making, offering an integrated interface to sort, repair, donate, or recycle any consumer product.

Step-by-Step User Flow

- 1. **Image Upload:** The user uploads a photo or captures an image of an item (e.g., a laptop or jacket).
- 2. **Computer Vision Processing:** The AI model classifies the item category, detects materials, and assesses its condition.
- 3. **Intelligent Routing:** The engine calculates scores for repair, donation, and recycling, recommending the best action path.
- 4. **Logistics & Impact:** The platform connects the user with nearby partners (NGOs or repair shops) and displays their personal carbon offset metrics.

System Interaction Flowchart



HERO SECTION & SIDE NAV

The landing page of **SecondLife** is designed to capture attention and communicate the platform's environmental mission. It features an interactive layout with premium visual feedback:

Dynamic 3D Particle Canvas

Behind the text layers, a WebGL-based Three.js particle canvas renders a rotating digital globe. As users scroll, the particles dynamically react to scroll speeds, forming orbits that symbolize materials moving through a circular economy.

Interactive Custom Cursor

A custom SVG-based cursor tracker follows the pointer, transforming into an emerald-tinted neon ring with a cyan center dot when hovering over interactive elements. This visual feedback makes browsing the site feel fluid and responsive.

Floating Glassmorphic Sidebar

A vertical navigation bar on the right side of the screen shows dots for each section of the page. It updates dynamically using a custom scroll listener, highlighting the current section as the user scrolls.

LANDING UI LAYOUT OVERVIEW

[Floating Header] → [ThreeJS Globe] → [Floating Audio Toggle]

AI SCANNER & HOW IT WORKS

The **AI Second Life Scanner** is the core functional area of the application. It consists of a split-screen dashboard that guides users through the item analysis pipeline:

Split-Screen Diagnostic Interface

The left column features a drag-and-drop file upload zone that triggers a scanning laser animation when an image is loaded. The right column shows real-time metrics, using radial progress bars to display repair, donation, and recycling scores.

Active Scanning Sound Design

During scanning, a synthesizer generates a digital hum that changes pitch using an LFO. Once the analysis completes, the hum fades out and plays an ascending C-major arpeggio chime, giving clear feedback to the user.

Interactive Timeline Loop



Step 01: Capture & Upload

User uploads an item photo or selects a quick-scan demo card.



Step 02: AI Material Identification

Neural networks classify materials and evaluate item wear.



Step 03: Direct Partner Action

Connect with local NGOs or schedule zero-emission item pickups.

DASHBOARD & COMMUNITY HUB

Beyond item scanning, SecondLife provides sections that encourage long-term sustainability habits and build community engagement:

ESG Impact Dashboard

The Dashboard shows personal and community sustainability metrics, tracking total carbon offsets, water conserved, and items diverted from landfills. Data grids render carbon savings using live tracking statistics.

Interactive Global Partner Map

A vector map shows active recycling centers, partner NGOs, and local repair shops. Users can click pins to see address details and coordinate donation collections.

Social Hub & Contact Form

The Social Hub shows real-world impact stories, educational resources on circularity, and a glassmorphic contact form for organizations wanting to join the SecondLife network.

ESG IMPACT PANEL

[Carbon Tracker] → [Partner Pins Map] → [Rewards Dashboard]

AI SCANNER & DONATION ENGINE

AI Object Identification & Wear Detection

The core scanning engine uses computer vision to classify items and assess their wear. It identifies materials and evaluates condition factors like battery health, surface wear, and fabric quality.

```
classifier.predict(image) → { class: "Electronics", wear: "Minor" }
```

Intelligent Donation Routing

Items with high donation scores (good condition, high utility) are matched with regional partners based on coordinates. This ensures electronics go to digital literacy programs and warm garments go to shelter networks.

Integrating location APIs allows donors to find and coordinate drop-offs with nearby charities with minimal travel distance.

By automating sorting and matchmaking, the platform simplifies donation logistics. This reduces manual work for NGOs and helps ensure donated items match their current needs.

REPAIR NETWORK & CARBON CALCULATOR

To support repair efforts, SecondLife connects users with local technicians and provides an automated calculator to show their environmental savings:

Localized Repair Marketplace

When the AI detects minor repairable faults, it lists certified local technicians. The page displays the technician's location, repair rating, and cost estimates, encouraging users to fix items rather than replacing them.

ESG Calculator Formula

The system calculates CO2 offsets and water conserved by comparing the impact of reusing an item against manufacturing a new one. The formula factors in manufacturing and transportation emissions:

$$E_{\text{saved}} = E_{\text{manufacturing}} \times (\text{Score} / 100) - E_{\text{transportation}}$$

Transparent Environmental Metrics

The carbon calculator uses data from life-cycle assessments (LCA) to show real-world metrics, like liters of water saved and kilograms of CO2 avoided, giving users clear feedback on their impact.

REWARDS PROGRAM & EDUCATION HUB

We use gamification and educational resources to encourage users to adopt circular economy habits over the long term:

Gamified Eco-Points & Leaderboards

Every time a user scans and successfully processes an item, they earn Eco-Points. The community dashboard displays active leaderboards, letting users see how they rank compared to others and motivating them to contribute more.

Localized Volunteer Networks

The platform includes a coordination portal for volunteer networks. Active users can sign up as eco-couriers to collect items from donors and deliver them to NGOs, earning extra rewards points.

Circularity Education Hub

The Education Hub offers guides and articles on upcycling, repair basics, and waste classification. This content teaches users how to make sustainable choices and manage resources more effectively.

These features transform ecological actions from a chore into a shared community activity, helping to build long-term habits.

TECHNOLOGY STACK

SecondLife is built using modern web technologies to deliver a fast, responsive, and engaging user interface:

Frontend Framework

Built on **React** and **Next.js** to handle server-side rendering, routing, and component states.

Next.js 16

React 19

TypeScript

Interactive Styling

Styled with **Tailwind CSS** for layout responsiveness, custom grid systems, and glassmorphic panels.

Tailwind CSS 4

Lucide Icons

Vanilla CSS

3D & Motion Engines

Uses **Three.js** and **Framer Motion** for smooth transitions, layout changes, and interactive 3D particle globes.

Three.js

R3F

Framer Motion

Lenis

Audio & AI Concepts

Features a **Web Audio API** synthesizer for interactive sound design and mock classification logic.

Web Audio API

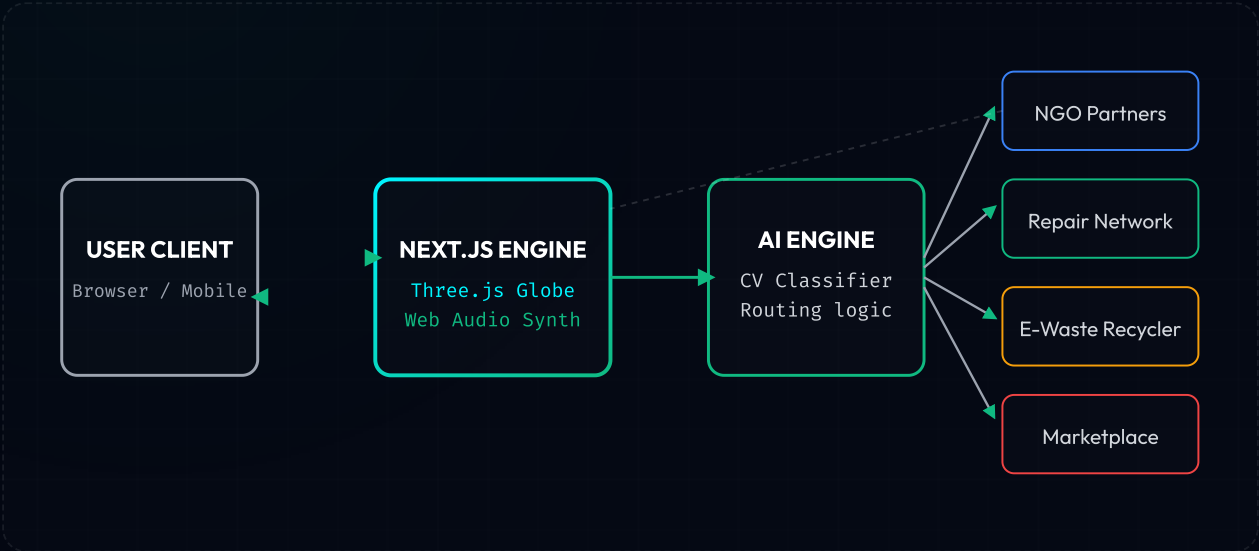
Mock Inference

Vercel Edge

This modern architecture ensures the platform loads quickly, runs smoothly on mobile devices, and handles interactive elements without lag.

SYSTEM ARCHITECTURE

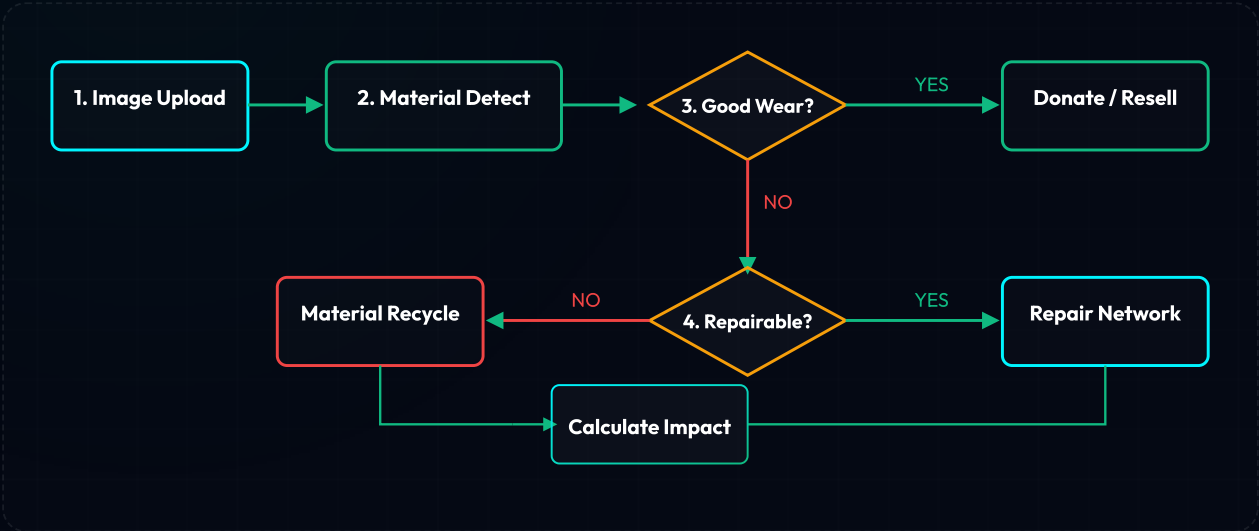
SecondLife uses a client-edge architecture designed to process user actions quickly and coordinate efficiently with local partners:



This structure ensures the user interface remains responsive on the client side, while heavy processing tasks are routed to specialized edge services for fast responses.

AI RECOMMENDATION WORKFLOW

The circular recommendation engine processes item data through a series of logical steps to determine the best environmental action path:



This logic flow ensures every scanned item is routed to the most sustainable option. Reusable goods go to donors, repairable ones to technicians, and worn-out items are directed to verified recycling facilities.

UI/UX DESIGN SYSTEM

The visual identity of **SecondLife** is designed to look like a premium tech platform rather than a simple class assignment. We use design tokens to maintain visual consistency across all sections:

Premium Neon & Space Dark Palette

The interface uses a very dark blue-grey background (`#030712`) to save screen energy. We use glowing **Emerald Green** (`#10B981`) and **Futuristic Cyan** (`#00F5FF`) as accent colors to guide attention to key interactive features.

Bg: #030712

Cyan: #00F5FF

Emerald: #10B981

Glassmorphism UI Patterns

Content is organized in cards with semi-transparent backgrounds and thin, light borders (`rgba(255,255,255,0.08)`). Applying CSS backdrop filters creates a clear hierarchy and keeps layout overlays looking clean.

Interactions & Micro-Feedback

Interactive elements scale up slightly on hover and compress on click. We use Web Audio API chimes and cursor changes to make actions like file uploads and button presses feel tactile and responsive.

SUSTAINABILITY IMPACT

We track several key environmental metrics to show the tangible impact of donating and recycling through SecondLife:

4,250 kg

Carbon Saved (CO2)

Diverting electronics and metals from landfills avoids the emissions required to mine and process raw materials.

48,500 L

Water Preserved

Extending the lifespan of textiles and garments prevents the water waste associated with manufacturing new fabrics.

380 units

Landfill Diversion

Bulky furniture, plastics, and appliances are successfully diverted into circular reuse cycles.

42 NGO

Partners Connected

A unified logistics interface connects donors directly with nearby humanitarian and shelter networks.

By converting recycling actions into clear, quantified carbon offsets, SecondLife provides transparent feedback that motivates users to make sustainable habits a permanent part of their routine.

ADVANTAGES & SCALABILITY

SecondLife combines a modern user interface with automated backend routing. This design offers several advantages over traditional recycling platforms:

Zero-Friction User Experience

Instead of manually filling out long forms, users can simply upload a photo of an item. The AI classifies it and recommends the best disposal path, reducing the effort needed to adopt sustainable habits.

Local Enterprise Integration

The platform connects users directly with local repair shops, NGOs, and recyclers. This decentralized approach creates a shared regional network, helping local businesses grow while keeping logistics costs low.

Scalable Platform Architecture

The modular React and Next.js frontend is designed to support expanding services, such as live AI image feeds, blockchain tracking, and carbon credit integrations, without requiring a complete rewrite of the system.

This combination of automated classification, local business integration, and scalable code makes SecondLife a viable foundation for a modern green-tech startup.

CHALLENGES FACED & MITIGATIONS

During the development of SecondLife, we addressed several technical challenges related to 3D rendering, browser audio constraints, and layout design:

1. WebGL & 3D Rendering in Next.js SSR

Importing heavy WebGL canvas components directly into Next.js triggered Server-Side Rendering (SSR) reference errors because the browser window object was not available on the server.

Mitigation: Used `next/dynamic` with `{ ssr: false }` to delay loading the `Three.js` globe component until the client-side browser is fully initialized.

2. Browser Audio Block Policy

Modern web browsers block synthesized `AudioContext` playback before a user interacts with the page, which initially caused sound effects to fail on first load.

Mitigation: Implemented global click and touch event listeners in `SoundManager.tsx`. These listeners automatically resume the `AudioContext` on the user's first interaction.

3. Real-Time Responsive Layout Grid

Maintaining glassmorphic styling, background particles, and text readability across different screen sizes was challenging. We solved this by using custom CSS backdrop-filters, modern container queries, and Tailwind responsive grid utilities.

FUTURE ENHANCEMENTS & ROADMAP

We plan to expand SecondLife with new features to improve item tracking, automate drop-offs, and increase user engagement:

AI Assistant Integration

We will integrate a conversational AI helper (like Gemini API) to answer user questions about materials, repair basics, and recycling options, guiding them through circular habits.

Blockchain-Based Donation Tracking

We plan to implement a decentralized public ledger to track donated items from drop-off to delivery, providing full transparency that helps build trust with donors and partner organizations.

IoT Smart Bins & Carbon Credits

We plan to deploy smart collection bins equipped with sensors to verify item drop-offs automatically, allowing users to earn carbon credits that can be traded on green marketplaces.

These additions will make the platform more automated, transparent, and connected to global green initiatives.

LEARNING OUTCOMES & CONCLUSION

Key Learning Outcomes

Building SecondLife provided practical experience with several modern web development and system design concepts:

- **Client-Side Audio Synthesis:** Learned to generate custom haptic sound cues using the Web Audio API without relying on external media files.
- **WebGL Integration:** Experienced managing Three.js canvas lifecycles and animations in a Next.js framework.
- **State & Event Listeners:** Structured global event listeners to handle user interactions cleanly while avoiding component memory leaks.

Conclusion

SecondLife demonstrates how modern web interfaces and AI concepts can be combined to address real-world environmental challenges. The application simplifies circular economy decisions by automating item classification and connecting users with local partners.

By combining interactive 3D elements, custom audio feedback, and data-driven impact metrics, the platform shows that ecological responsibility can be made both visually engaging and easy to adopt. It provides a functional foundation for scalable green-tech initiatives.

PROJECT LINKS & QR CODES

Scan the QR codes below to access the open-source repository or visit the live deployment of the platform:

GitHub Repository



`melvindeepesh-boop/SECOND-LIFE`

Open-Source Code

Live Website



`second-kohl-ten.vercel.app`

Interactive Demo

Scan with any mobile device to view details, inspect build pipelines, or run live scans.

REFERENCES

Below are the references, standards, and technical documentation used during the design and development of the SecondLife platform:

1. Circular Economy & Sustainability

- * Ellen MacArthur Foundation. (2020). *Towards the Circular Economy: Economic and business rationale for an accelerated transition*.
- * United Nations. (2015). *Sustainable Development Goal 12: Responsible Consumption and Production*.
- * Ghisellini, P. et al. (2016). *A review on circular economy: the expected transition to a balanced interplay of environmental and economic systems*. Journal of Cleaner Production.

2. Web Graphics & WebGL Concepts

- * Cabello, R. (2025). *Three.js Documentation and Examples*. WebGL library resource.
- * React Three Fiber (R3F) Core Team. *Declarative 3D scenes in React framework documentation*.
- * Parisi, T. (2014). *Learning Three.js: The JavaScript 3D Library for WebGL*. O'Reilly Media.

3. Web Audio & Frontend Frameworks

- * W3C Web Audio Working Group. (2021). *Web Audio API Specification*.
- * Next.js Core Team. (2026). *Next.js App Router Architecture and Server Components Guide*. Vercel docs.
- * Framer Motion Team. *Layout animations and physics-based motion in React*.